Lab 27 — File Integrity Monitoring with AIDE

You will baseline critical files with AIDE, modify one, and watch the diff report.

Run on https://killercoda.com/playgrounds/scenario/ubuntu

Required software (free, apt): - `aide`

Step 1 — Install

```
apt update && apt install -y aide
```

```
aide --version
```

Step 2 — Configure scope

```
cat > /etc/aide/aide.conf.d/99_lab <<'EOF'
```

```
# Debian/Ubuntu AIDE ships no LSPP/NORMAL macro, so define our own
check groups
```

```
LAB_NORMAL = p+i+n+u+g+s+m+c+sha256
```

```
LAB_STRICT = p+i+n+u+g+s+b+m+c+acl+xattrs+sha256+sha512

/etc/passwd  LAB_NORMAL

/etc/shadow  LAB_NORMAL

/etc/sudoers LAB_NORMAL

/usr/bin     LAB_STRICT

/usr/sbin    LAB_STRICT

EOF
```

`LAB_NORMAL` checks size, perms, ownership and a SHA-256 hash. `LAB_STRICT` adds ACLs and extended attributes for the strictest coverage.

Step 3 — Initialise the baseline

```
aideinit -y -f

mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db
```

```
ls -lh /var/lib/aide/aide.db
```

The baseline is a SHA-256 fingerprint database. **Treat it like a key — back it up off-host.**

Step 4 — First check (should report no changes)

```
aide --check | head -10
```

Step 5 — Tamper and re-check

```
echo "# tampered $(date)" >> /etc/sudoers

useradd -m attacker

aide --check 2>&1 | head -25
```

AIDE reports the changed sudoers and the new shadow entry — exactly the signal a SOC needs.

Step 6 — Roll the baseline forward (legitimate change)

After approved patches:

```
aide --update -B 'database_new=file:/var/lib/aide/aide.db.new'
```

```
mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db
```

Step 7 — Schedule a daily check

```
cat > /etc/cron.daily/aide-check <<'EOF'
#!/bin/sh
/usr/bin/aide --check | logger -t aide
EOF
chmod +x /etc/cron.daily/aide-check
```

Logs go to syslog → SIEM.

Free online tools

- **AIDE docs** — https://aide.github.io
- **Tripwire Open Source** — https://github.com/Tripwire/tripwire-open-source
- **Samhain** — https://www.la-samhna.de/samhain
- **Wazuh FIM** — https://documentation.wazuh.com/current/user-manual/capabilities/file-integrity/index.html

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What you learned

- A cryptographic baseline is the only reliable way to spot file tampering.
- The two operational mistakes: storing the DB on the host, never updating after patching.
- How to wire AIDE into syslog for SIEM correlation.